Associate Software Engineer – Take-Home Assignment

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How this works

- **AI tools are allowed, and we encourage them,** use whatever you like. We hire people who get a lot out of AI.

The business problem

We run a D2C haircare brand that sells on our own website (Shopify). About **55% of our orders are Cash on Delivery (COD)**.

The problem: around **28% of COD orders are refused at the door or never delivered**. This is called RTO (Return to Origin). For every RTO we lose:

- Forward shipping: ₹65
- Return shipping: ₹65
- Packaging + handling: ₹25
- Around 8% of returned products come back damaged and can't be resold (average product cost ₹180)

Prepaid orders have an RTO rate of only about 4%.

We ship about **3,000 orders a month**. Average order value is ₹649. Our gross margin on a delivered order is about 62%.

We want an AI-powered system that looks at every COD order **before it is shipped** and decides what to do with it.

What two of our leaders are saying

The two people you'd work with disagree. Read both carefully.

Head of Operations:

> "RTO is killing us. I want any risky COD order cancelled automatically. If a pincode has bad history, just stop COD there. I'd rather lose a few sales than keep burning money on returns."

Head of Marketing:

> "We spend ₹300+ to acquire each new customer through Meta ads. If we cancel their first order, we've wasted that money and they'll never come back. Most first-time buyers choose COD because they don't trust us yet. Please don't kill growth to save shipping costs."

You don't have to pick a side. But your solution has to deal with both concerns, and you must explain your reasoning in money (₹), not opinions.

What you need to build

Part 1 — The decision engine (core)

Using the dataset we share (`orders_sample.csv`, about 2,000 past COD orders with whether they were delivered or returned), build a system that takes a new COD order and outputs one of these actions:

1. **Ship as-is**
2. **Ship after confirmation** (customer must confirm on WhatsApp first)
3. **Nudge to prepaid** (offer a small discount to pay online)
4. **Hold / cancel**

Requirements:

- Use an LLM where it actually helps (for example: judging whether an address is complete and real, spotting fake names like "asdf" or "test", reading Hinglish addresses). **Do not use an LLM for things simple code or maths can do better.** Tell us where you drew that line and why.
- Every decision must come with a short, plain-English reason that an operations person can understand.
- The thresholds (when to cancel vs. confirm vs. nudge) must be easy for a non-technical person to change, without touching code.

Part 2 — The WhatsApp confirmation flow

Design (and build a working mock of) the message flow for "Ship after confirmation" and "Nudge to prepaid":

- What message goes out, in what language, and when
- What happens if the customer replies "yes", "no", something random, or doesn't reply at all in 24 hours

- How much discount you'd offer for prepaid, and **why that number** (show the maths)

You don't need a real WhatsApp account. A mock API or a simple chat simulator is fine.

Part 3 — A simple dashboard

One screen our founder can look at every morning. It should answer:

- How many orders were in each of the 4 buckets yesterday
- Estimated ₹ saved from avoided RTOs
- Estimated ₹ lost from cancelled orders that might have been genuine
- **Net ₹ impact**

Part 4 — The business memo (most important part)

A 1–2 page memo written to the founder. Not a technical document. It should cover:

1. What you found in the data that surprised you (at least 3 real findings, with numbers)
2. Your recommended rules, and what they would have done to last month's orders
3. How you balanced Operations vs. Marketing, with the ₹ trade-off
4. What could go wrong with your system and how we'd notice
5. What you'd build next if you had 2 more weeks

What to submit

1. **GitHub repo** (public or give us access) with a README that lets us run it in under 10 minutes
2. **The business memo** (PDF or Google Doc)
3. **Decision log** — a simple running list of the important choices you made, what alternatives you considered, and why you picked what you did. Bullet points are fine. This matters more than polish.
4. **AI usage log** — which AI tools you used, for what, and 2–3 examples where the AI gave you something wrong or weak and how you caught and fixed i